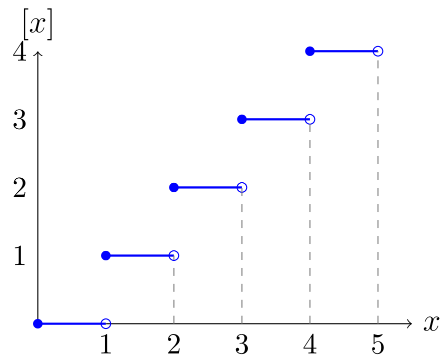
# Explainer: Problem 1 (Rudin 6.16) — The Zeta Function via Integrals

## What's going on?

The Riemann zeta function $\zeta(s) = \sum_{n=1}^{\infty} 1/n^s$ is a sum over integers. The problem asks us to express it as an integral. The trick is that the *floor function* $[x]$ is piecewise constant (it equals $n$ on $[n, n+1)$), so integrals involving $[x]$ decompose into sums.

## The floor function




On the interval $[n, n+1)$, $[x] = n$. So an integral like $\int_1^\infty \frac{[x]}{x^{s+1}} dx$ becomes a sum of pieces, each of which is easy: $\int_a^b x^{-s-1} dx$.

## Strategy for (a)

1. Break $[1, \infty)$ into $[n-1, n)$ pieces where $[x] = n - 1$.
2. Evaluate each piece: $\int_{n-1}^{n} (n-1)x^{-s-1} dx$ gives a clean expression.
3. Sum up. The resulting series **telescopes** — the coefficient of each $1/k^s$ collapses to 1, recovering $\zeta(s)$.

**Telescoping intuition:** When you have $\sum (n-1)(a_{n-1} - a_n)$, rewriting by collecting coefficients of each $a_k$ gives $[k - (k-1)] \cdot a_k = a_k$. That's the same trick as summation by parts.

## Strategy for (b)

Just write $x = [x] + (x - [x])$ and split the integral. The $[x]$ piece is $\zeta(s)/s$ by (a); the $x$ piece is an elementary integral giving $1/(s-1)$. Rearranging gives (b).

## Convergence of the integral in (b) for $s > 0$

Use that $0 \leq x - [x] < 1$, so the integrand is dominated by $1/x^{s+1}$, which has a finite integral on $[1, \infty)$ whenever $s > 0$. This is significant because the *series* defining $\zeta(s)$ only converges for $s > 1$ — formula (b) gives a way to extend $\zeta$ beyond $s = 1$.